

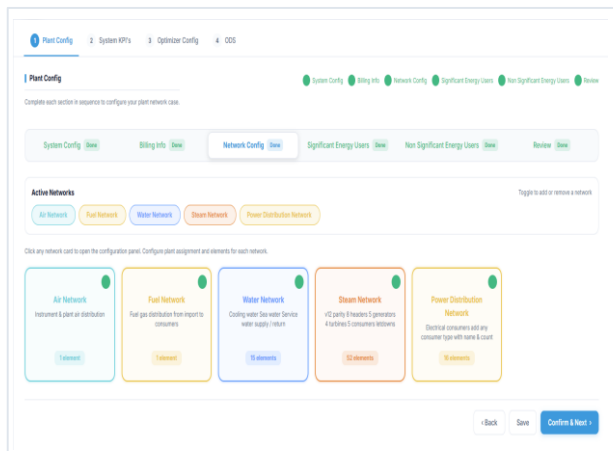
PRODUCT OVERVIEW

EnergyX360 is a real-time decision-support solution that identifies the most energy-efficient operating strategy for a process plant. It uses real-time plant data, process engineering models, equipment operating constraints and MILP-based optimization to determine the best combination of boilers, turbines, motors, compressors and utility systems that satisfies plant demand. Optimization opportunities are converted into quantified, actionable recommendations for the process and operations teams, configured for each plant's utility set-up, equipment limits and operating philosophy.



01 Plant & Utility Network Configuration

Shown: plant utility configuration, equipment relationships



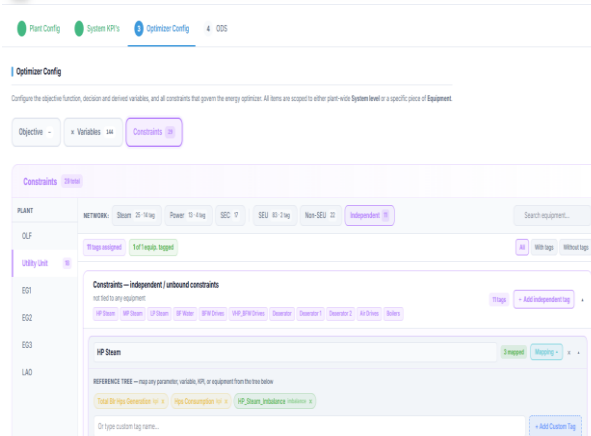
Steam System
Boiler System
Rotating Equip.
Utility Systems

HP/MP/LP steam headers, generation, consumption, venting and dumping
boiler operating limits, minimum/maximum loads, fuel consumption and steam generation characteristics
turbine- and motor-driven pumps/compressors with capacities, limits and changeover options
BFW, DMW, condensate, deaerator, cooling water and compressed air; linked to process-unit demand



02 MINLP Optimization Engine

Shown: configured plant utility system, real-time operating condition and optimisation workflow.



Inputs
Processing
Outputs

Define Objective Function, Constraints, Variables, equipment status, process loads and utility demands
Tag validation, operating-condition identification.
MILP-optimized targets, equipment switchover recommendations, utility allocation and savings



03 Optimization Modules

A site enables only those optimisation modules supported by its assets, operating philosophy and available data.



Steam System Optimization
Turbine-Motor Drive Switchover
Boiler System Optimization
Air Compressor System Optimization

Optimizes HP/MP/LP steam generation and consumption while minimizing excess dumping and venting
Evaluates turbine and motor-driven equipment and recommends switchovers to reduce steam consumption
Optimizes boiler loading and steam generation to reduce fuel consumption within operating constraints
Optimizes compressor loading and IGV operation to reduce excess air venting

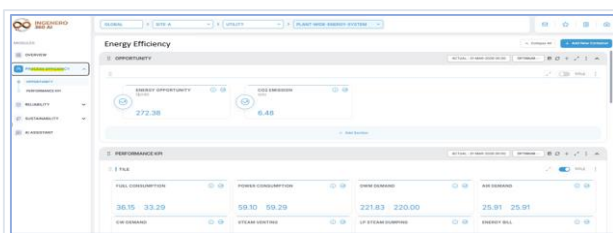


BFW & DMW Optimization
Cooling Water Optimization

Optimizes BFW generation, condensate return, DMW makeup and deaerator steam requirements for new steam demand
Optimises the number and type of CW drives in operation based on actual demand



04 Optimization Opportunities & Operator Actions



EFFECT MESSAGE	CAUSE MESSAGE	ACTUAL	OPTIMUM	SUGGESTION	ACTIONS
POWER BILL	High consumption due to high decrease	1.33	0.57	INCREASE THE SETPOINT OF VENT OFF VALVE BY 0.05 BARs, WHICH WILL HELP REDUCE AIR VENTING	IF
SYSTEM - FUEL CONSUMPTION	BFW C TURBINE OVERBURN	1.00	0.00	SWITCH OFF BFW C TURBINE TO REDUCE DUMP/VENT OF STEAM	IF
	DMW TURBINE A OVERBURN	1.00	0.00	SWITCH OFF DMW TURBINE A TO REDUCE DUMP/VENT OF STEAM	IF
	DEAERATORS A/B	84.80	79.95	OPTIMIZED HP STEAM GENERATOR WILL REDUCE THE BFW CONSUMPTION. EVENTUALLY IT WILL REDUCE LP STEAM DEMAND IN DE-DEAERATOR. REFER OPTIMIZATION RESULTS FOR DETAILS	IF

2-3%

Energy Bill Reduction
from compressor, BFW, Steam turbines and cooling-water network optimization

3-4%

CO2 Emissions Reduction
net reduction across all optimization actions

3-5%

Energy Saving
Steam and Power Savings

Operator Actions: Turbine → Motor changeover, Optimized boiler loading, cooling-water pump and air compressor drive optimization and Steam Allocation for Actual Steam Demand.



05 Platform & Deployment

Historian Connectivity
connects to PI/DCS data for continuous monitoring

Optimisation Engine
MILP-based optimisation with what-if analysis of changes

Operator Dashboard
current conditions, recommendations, impact and savings

Daily Reporting
tracks performance, actions recommended, taken and realised savings

Scalable Deployment
reusable framework adaptable to different plants and configurations

DELIVERY

Configured, validated and commissioned by Ingenero process engineers.

FLEXIBILITY

Any utility configuration; selectable optimisation modules; configurable constraints

SCALE

Repeat sites reuse the optimisation framework, with no/minimal rework per plant.

Request a customised demo
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